

Mathematical Model of Drone Return Behavior

1 Vector Toward Home

Let current position be:

$$(x, y)$$

Home position:

$$(x_h, y_h)$$

Direction components

$$dx = x_h - x$$

$$dy = y_h - y$$

Distance to home

$$d_h = \sqrt{dx^2 + dy^2}$$

This is the Euclidean distance. It defines how far the drone is from home.

2 Speed Constraint Model

Let:

Δt = time step

Maximum allowed speed:

Normal return:

$$v_{cap} = V_{max}$$

Degraded return:

$$v_{cap} = 0.55V_{max}$$

Emergency return:

$$v_{cap} = 0.75V_{max}$$

Actual speed per update

$$v = \min \left(v_{cap}, \frac{d_h}{\Delta t} \right)$$

Meaning:

- If far $\rightarrow$ move at capped speed.
 - If close $\rightarrow$ slow down so it does not overshoot.
-

3 Direction Normalization

Unit vector toward home:

$$\hat{u}_x = \frac{dx}{d_h}$$
$$\hat{u}_y = \frac{dy}{d_h}$$

This ensures motion purely in the home direction.

4 Position Update (Core Kinematics)

$$x_{new} = x + \hat{u}_x \cdot v\Delta t$$
$$y_{new} = y + \hat{u}_y \cdot v\Delta t$$

This is first-order discrete-time motion:

$$\vec{p}_{t+1} = \vec{p}_t + \vec{v}\Delta t$$

where:

$$\vec{v} = v\hat{u}$$

5 Wind Disturbance Model (Degraded Mode)

Wind phase evolution:

$$\phi_{t+1} = \phi_t + 0.7\Delta t$$

This creates a time-varying oscillation.

Disturbed wind components

$$\begin{aligned}g_x &= w_x(0.65 + 0.35\sin \phi) \\g_y &= w_y(0.65 + 0.35\cos (0.9\phi))\end{aligned}$$

This makes wind amplitude oscillate between:

$$0.65 - 0.35 = 0.30$$

and

$$0.65 + 0.35 = 1.00$$

So wind strength varies between 30% and 100% of base wind.

6 Partial Wind Compensation

Compensation factor:

$$c = 0.45$$

Residual disturbance:

$$1 - c = 0.55$$

Applied disturbance:

$$\begin{aligned}x_{new} &= x + g_x \cdot 0.55\Delta t \\y_{new} &= y + g_y \cdot 0.55\Delta t\end{aligned}$$

So only 55% of wind affects the drone.

7 Combined Position Update (Degraded Mode)

Full discrete update:

$$\begin{aligned}x_{new} &= x + \hat{u}_x v \Delta t + 0.55 g_x \Delta t \\y_{new} &= y + \hat{u}_y v \Delta t + 0.55 g_y \Delta t\end{aligned}$$

This is:

$$\vec{p}_{t+1} = \vec{p}_t + \vec{v}_{home} \Delta t + \vec{v}_{wind} \Delta t$$

8 Altitude Descent Model

Let altitude:

$$z$$

Home altitude:

$$z_h$$

Update rule:

$$z_{new} = \max(z_h, z - r_d \Delta t)$$

Descent rate

Normal return:

$$r_d = 0.5V_{land}$$

Degraded return:

$$r_d = 0.6 \times 0.5V_{land}$$

$$r_d = 0.3V_{land}$$

So degraded descent is slower for stability.

9 Wind Initialization

Initial wind:

$$w_x = 1.2 \cos(\phi_0)$$

$$w_y = 1.2 \sin(\phi_0)$$

Magnitude:

$$\sqrt{w_x^2 + w_y^2} = 1.2$$

So each drone has identical wind strength but different phase orientation.

10 System Nature (Mathematical View)

The motion equation is a discrete-time dynamical system:

$$\vec{p}_{t+1} = \vec{p}_t + \vec{f}_{deterministic} + \vec{f}_{disturbance}$$

Where:

- Deterministic term = controlled return vector
- Disturbance term = partially compensated oscillatory wind

This creates:

- Stable convergence toward home
- Reduced speed under degradation
- Controlled descent
- Realistic lateral disturbance

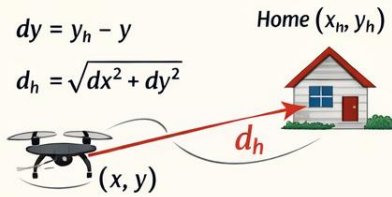
Degraded Return Navigation

1 Vector to Home

$$dx = x_h - x$$

$$dy = y_h - y$$

$$d_h = \sqrt{dx^2 + dy^2}$$



2 Speed Cap Limit

v_{max} Normal

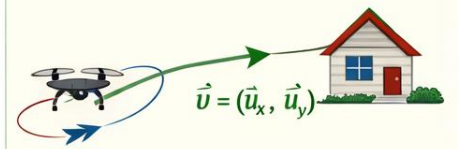
$0.55v_{max}$ Degraded

$0.75v_{max}$ Emergency

$$v = \min(v_{cap}, \frac{d_h}{\Delta t})$$

3 Direction Vector

$$\hat{u}_x = \frac{\hat{d}_x}{d_h} - \frac{\hat{d}_y}{d_h}$$



4 Wind Disturbance

$$\phi = \phi + 0.7\Delta t$$

$$g_x = w_x (0.65 + 0.35\sin\phi)$$

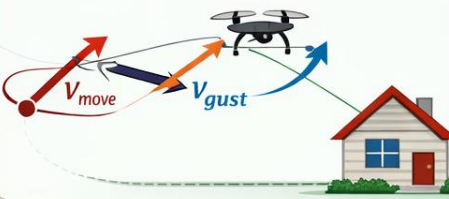
$$g_y = w_y (0.65 + 0.35\cos 0.9\phi)$$



3 Position Update

$$x_{new} = x + \hat{u}_x v \Delta t + 0.55 g_x \Delta t$$

$$y_{new} = y + \hat{u}_y v \Delta t + 0.55 g_y \Delta t$$

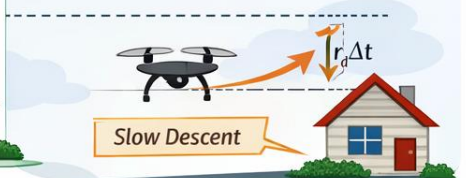


4 Altitude Descent

$$Z_{new} = \max(Z_h, z - r_d \Delta t)$$

Normal: $r_d = 0.5 v_{land}$

Degraded: $r_d = 0.3 v_{land}$



Reach Home

Emergency Landing

Safely Land at Home